11519 - Digging to the Photosphere: Circumplanetary Disk Emission from the Gap-Opening Protoplanet WISPIT 2 b

Cycle: 5, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
Dr. Haochang Jiang (PI) (ESA Member)	**Max Planck Institute for Astronomy**
Dr. Matthias Samland (CoI) (ESA Member) (CoPI)	Max Planck Institute for Astronomy

OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
WISPIT-2 imaging				
	1	WISPIT-2 Background	MIRI Imaging	(2) WISPIT-2-bkg
	2	WISPIT-2 roll1	MIRI Imaging	(1) WISPIT-2
	3	WISPIT-2 roll2	MIRI Imaging	(1) WISPIT-2
Reference Star imaging				
	4	Reference Star	MIRI Imaging	(5) WISPIT-2-ref

ABSTRACT

Giant planets build their mass by accreting material from circumplanetary disks. Studying this growth phase through direct imaging is challenging because it requires high angular resolution, extreme contrast, and specific wavelength coverage. In the near-IR, planetary photospheres dominate the emission. Only in the mid-infrared does the circumplanetary disk begin to outshine the planet—precisely the territory of JWST/MIRI.

Recently, a protoplanet was imaged inside the gap of the young T Tauri star WISPIT 2. Current detections all fall in near IR, resembling a planetary photosphere with an effective temperature of ~1000 to 1200 K. According to the WISE photometry, its host star is remarkably faint in the mid IR, offering a optimal contrast range to even detect the photosphere of WISPIT 2 b at the mid IR.

We propose mapping the SED of WISPIT 2 b from 5 to 15 micros through MIRI imaging with 5 filters. This simple yet powerful dataset will enable to characterize the physical properties of any detected circumplanetary disks. If a mid-IR excess is confirmed, WISPIT 2 b would represent only the second direct detection of a circumplanetary disk embedded within its natal disk—after PDS 70 c—and the first observed during the gap-opening phase. Even without a CPD excess, the program will yield the first direct mid-infrared detection of the photosphere of a protoplanet, which is a unprecedented milestone in itself.

This program will not only advance planet and moon formation studies, but also demonstrate the capabilities and limitations of MIRI non-coronagraphic high-contrast imaging.

OBSERVING DESCRIPTION

We request MIRI non-coronagraphic imaging observations to image the circumplanetary disk (CPD) around the young protoplanet WISPIT 2 b, which is located inside a shallow gas gap of its natal disk.

We will use the F560W, F770W, F1000W, F1280W, and F1500W filters to constrain the SED of the protoplanet+CPD at these wavelengths for the first time.

The exposure time necessary to achieve our goals was calculated using the most recent JWST exposure time calculator (ETC v5.0) and we used the Astronomer's Proposal Tool (APT v.2025.5.3); we simulated our target using the measured SED of the central star by compiling all available archival phometry. The protoplanet flux of planet was estimated using a very conservative assumption of a pure blackbody.

Our target will not saturate in any of the aforementioned filters in the SUB64 subarray. We first take dedicated background images for all filters to remove detector artifacts and extended background emission. We will then take the science target at two roll angles ~10 degree apart. This is followed by the observation of two reference stars. Only very low level mid-IR excess around the central star appear in the requested wavelength range, we will use a nearby star as reference for subtracting the point-spread function (PSF) using reference star differential imaging (RDI).

Using the SearchCal tool, we have selected an appropriate reference star: the evolved star HD 183267. We have checked the reference star for variability, binarity, and nearby stellar contaminants.

The on-target integration time is 60 minutes per filter (split over two angles). The associate background observation are the same length, as are the reference star observations (30 min per filter, without roll angle change). The total time required to execute the program is 15.6 hours.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	WISPIT-2	RA: 19 23 17.0332 (290.8209717d) Dec: -07 40 55.08 (-7.68197d) Equinox: J2000	Proper Motion RA: 6.308 mas/yr Proper Motion Dec: -27.137999904880417 mas/yr Parallax: 0.0074649" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[K stars, Protoplanetary disks]				
(2)	WISPIT-2-bkg	RA: 19 23 16.3000 (290.8179167d) Dec: -07 40 40.00 (-7.67778d) Equinox: J2000		
Comments: Category=Calibration Description=[Detector linearity test, Telescope/sky background]				
(5)	WISPIT-2-ref	RA: 19 28 58.8828 (292.2453450d) Dec: -06 10 25.62 (-6.17378d) Equinox: J2000	Proper Motion RA: -1.628 mas/yr Proper Motion Dec: -5.444000066745502 mas/yr Parallax: 0.0011903999999999999" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Calibration Description=[Point spread function]				

Observation

Proposal 11519, Observation 1: WISPIT-2 Background

Fri Mar 13 21:07:22 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	WISPIT-2-bkg	RA: 19 23 16.3000 (290.8179167d) Dec: -07 40 40.00 (-7.67778d) Equinox: J2000		

Comments:
Category=Calibration
Description=[Detector linearity test, Telescope/sky background]

Template

Subarray
SUB64

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	4-Point-Sets				1	1	POINT SOURCE	POSITIVE	DEFAULT

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F560W	FASTR1	9	520	1	Dither 1	4	2080	1770.156	269134
2	F770W	FASTR1	13	370	1	Dither 1	4	1480	1763.346	269134
3	F1000W	FASTR1	40	125	1	Dither 1	4	500	1744.62	269134
4	F1280W	FASTR1	40	125	1	Dither 1	4	500	1744.62	269134
5	F1500W	FASTR1	40	125	1	Dither 1	4	500	1744.62	269134

Observation

Proposal 11519, Observation 2: WISPIT-2 roll1

Fri Mar 13 21:07:22 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	WISPIT-2	RA: 19 23 17.0332 (290.8209717d) Dec: -07 40 55.08 (-7.68197d) Equinox: J2000	Proper Motion RA: 6.308 mas/yr Proper Motion Dec: -27.137999904880417 mas/yr Parallax: 0.0074649" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[K stars, Protoplanetary disks]

Template

Subarray
SUB64

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	4-Point-Sets				1	1	POINT SOURCE	POSITIVE	DEFAULT

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F560W	FASTR1	9	520	1	Dither 1	4	2080	1770.156	269134
2	F770W	FASTR1	13	370	1	Dither 1	4	1480	1763.346	269134
3	F1000W	FASTR1	40	125	1	Dither 1	4	500	1744.62	269134
4	F1280W	FASTR1	40	125	1	Dither 1	4	500	1744.62	269134
5	F1500W	FASTR1	40	125	1	Dither 1	4	500	1744.62	269134

Observation

Proposal 11519, Observation 3: WISPIT-2 roll2

Fri Mar 13 21:07:22 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	WISPIT-2	RA: 19 23 17.0332 (290.8209717d)	Proper Motion RA: 6.308 mas/yr	
		Dec: -07 40 55.08 (-7.68197d)	Proper Motion Dec: -27.137999904880417 mas/yr	
		Equinox: J2000	Parallax: 0.0074649"	
			Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[K stars, Protoplanetary disks]

Template

Subarray
SUB64

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	4-Point-Sets				1	1	POINT SOURCE	POSITIVE	DEFAULT

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F560W	FASTR1	9	520	1	Dither 1	4	2080	1770.156	269134
2	F770W	FASTR1	13	370	1	Dither 1	4	1480	1763.346	269134
3	F1000W	FASTR1	40	125	1	Dither 1	4	500	1744.62	269134
4	F1280W	FASTR1	40	125	1	Dither 1	4	500	1744.62	269134
5	F1500W	FASTR1	40	125	1	Dither 1	4	500	1744.62	269134

Observation

Proposal 11519, Observation 4: Reference Star

Fri Mar 13 21:07:22 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(5)	WISPIT-2-ref	RA: 19 28 58.8828 (292.2453450d) Dec: -06 10 25.62 (-6.17378d) Equinox: J2000	Proper Motion RA: -1.628 mas/yr Proper Motion Dec: -5.444000066745502 mas/yr Parallax: 0.0011903999999999999" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Calibration
Description=[Point spread function]

Template

Subarray
SUB64

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	4-Point-Sets				1	1	POINT SOURCE	POSITIVE	DEFAULT

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F560W	FASTR1	9	520	1	Dither 1	4	2080	1770.156	269134
2	F770W	FASTR1	13	370	1	Dither 1	4	1480	1763.346	269134
3	F1000W	FASTR1	40	125	1	Dither 1	4	500	1744.62	269134
4	F1280W	FASTR1	40	125	1	Dither 1	4	500	1744.62	269134
5	F1500W	FASTR1	40	125	1	Dither 1	4	500	1744.62	269134